HOMEWORK #4 SOLUTIONS

For the following problems, use μ_E=398,600.4418km^3/sec^2 and R_E=6,378km if necessary.

1. Prove that the eccentricity vector $\vec{e} = \frac{1}{\mu}\left(\dot{\vec{r}} \times \vec{h} - \frac{\mu}{r}\vec{r}\right)$ points to the periapsis.

   Hint: use the perifocal coordinate system.

Solution: show that $\vec{e}$ has only P-component in perifocal coordinate system(PQW).

$$\vec{r} = \begin{bmatrix} rcos(\nu) \\ rsin(\nu) \\ 0 \end{bmatrix}, \quad \vec{v} = \sqrt{\frac{\mu}{p}}\begin{bmatrix} -\sin(\nu) \\ e+\cos(\nu) \\ 0 \end{bmatrix}$$

$$\vec{e} = \frac{1}{\mu}\left(\vec{v} \times \vec{h} - \frac{\mu}{r}\vec{r}\right) = \frac{1}{\mu}\left[\left(v^2 - \frac{\mu}{r}\right)\vec{r} - (\vec{r}\cdot\vec{v})\vec{v}\right]$$

$$v^2 = \frac{\mu}{p}(e^2 + 2ecos(\nu) + 1), \qquad \vec{r}\cdot\vec{v} = r\sqrt{\frac{\mu}{p}}esin(\nu)$$

$$\vec{e}_Q = \left[\frac{\mu}{p}(e^2 + 2ecos(\nu) + 1) - \frac{\mu}{r}\right]r\sin(\nu) - r\sqrt{\frac{\mu}{p}}esin(\nu)\cdot\sqrt{\frac{\mu}{p}}(e + \cos(\nu))$$

$$= \frac{\mu}{p}(ecos(\nu) + 1)\cdot r\sin(\nu) - \mu\sin(\nu) = 0$$

$$\because r = \frac{p}{1 + e\cos(\nu)}$$

Similarly, it can be shown that $\vec{e}_P \neq 0$, hence, $\vec{e}$ is aligned with periapsis direction.

2. Rader readings determine that an object is located at

   $r = 1.2\hat{K}\ DU$ and $v = \left(0.4\hat{I} + 0.3\hat{K}\right)DU/TU$

   Determine p, e, I, Ω, ω, ν, u, λ_{true}.

Solution:

$$r = 1.2\ \mathrm{DU}$$

$$v = 0.5\ \mathrm{DU/TU}$$

$$\vec{e} = \frac{1}{\mu}\left[\left(v^2 - \frac{\mu}{r}\right)\vec{r} - (\vec{r}\cdot\vec{v})\vec{v}\right] = -0.1440\hat{I} - 0.8080\hat{K}$$

$$e = 0.8207$$

$$\xi = \frac{v^2}{2} - \frac{\mu}{r} = -0.7083\ \mathrm{DU^2/TU^2}$$

$$a = -\frac{\mu}{2\xi} = 0.7059\ \mathrm{DU}$$

$$P = a(1 - e^2) = 0.1626\ \mathrm{DU}$$

$$\vec{h} = \vec{r}\times\vec{v} = 0.48\hat{J},\ \ h = 0.48$$

$$\cos(i) = \frac{h_K}{|\vec{h}|} = 0,\ \ i = 90°$$

$$\vec{n} = \hat{K}\times\vec{h} = -0.48\hat{I},\ \ n = 0.48$$

$$\cos(\Omega) = \frac{n_I}{|\vec{n}|} = -1,\ \ \Omega = 180°$$

$$\cos(\omega) = \frac{\vec{n}\cdot\vec{e}}{|\vec{n}||\vec{e}|} = 0.1755,\ \ \omega = 79.8950°$$

$$\text{if } e_K < 0\ then\ \omega = 360° - \omega = 280.1050°$$

$$\cos(\nu) = \frac{\vec{e}\cdot\vec{r}}{|\vec{e}||\vec{r}|} = -0.9845,\ \ \nu = 169.8950°$$

$$\cos(\mathrm{u}) = \frac{\vec{\mathrm{n}} \cdot \vec{\mathrm{r}}}{|\vec{\mathrm{n}}||\vec{\mathrm{r}}|} = 0, \;\; \mathrm{u} = 90°$$

$$\cos(\,\lambda_{true}) = \frac{\vec{\mathrm{r}_1}}{|\vec{\mathrm{r}}|} = 0, \qquad \lambda_{true} = \Omega + \mathrm{u} = 270°$$

3. Given that the Mean Anomaly $M=235.4°$ and $e=0.4$, find the eccentric anomaly E.

Solution: Example 2-1 of Vallado

4. Given that the Mean Anomaly $M=235.4°$ and $e=2.4$, find the hyperbolic anomaly H.

Solution: Example 2-3 of Vallado

5. Given the coordinates $\Phi_{gd} = -7° \; 54' \; 23.886''$, $\lambda = 345° \; 35' 51.000''$ and h_{ellp}=56m, find the position vectors.

Solution: Example 3-2 of Vallado

6. Example 3-3 in Chapter 3 of Vallado (pp. 177-178): Given $\vec{r}_{IJK} = 6{,}524.834\hat{I} + 6{,}862.875\hat{J} + 6{,}448.296\hat{K}$ km on February 24, 1995, 12:00 UTC, find latitude and longitude.

Solution: Example 3-3 of Vallado

7. Find the Julian Date of October 26, 1996, 2:20PM UT.

Solution: Example 3-4 of Vallado

8. Find the GMST(Θ_{GMST}) and LST(Θ_{LST}) of August 20, 1992, 12:14PM at 104° west longitude.

Solution: Example 3-5 of Vallado

9. Example 3-7 in Chapter 3 of Vallado (pp. 197-198): Find UT1, TAI, TT, T_{UT1}, T_{TT}, T_{TDB} of May 14, 1990, 10:43 Mountain Standard Time (UTC based)

Solution: Example 3-7 of Vallado